

A Dynamic Model of the Central European Power Grid

for investigating system separations

Project Laboratory

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HALLGATÓI NYILATKOZAT

Alulírott Babai András, szigorló hallgató kijelentem, hogy ezt az önálló laboratóriumot meg nem engedett segítség nélkül, saját magam készítettem, csak a megadott forrásokat (szakirodalom, eszközök stb.) használtam fel. Minden olyan részt, melyet szó szerint, vagy azonos értelemben, de átfogalmazva más forrásból átvettem, egyértelműen, a forrás megadásával megjelöltem.

Hozzájárulok, hogy a jelen munkám alapadatait (szerző(k), cím, angol és magyar nyelvű tartalmi kivonat, készítés éve, konzulens(ek) neve) a BME VIK nyilvánosan hozzáférhető elektronikus formában, a munka teljes szövegét pedig az egyetem belső hálózatán keresztül (vagy hitelesített felhasználók számára) közzétegye. Kijelentem, hogy a benyújtott munka és annak elektronikus verziója megegyezik. Dékáni engedéllyel titkosított diplomatervek esetén a dolgozat szövege csak 3 év eltelte után válik hozzáférhetővé.

Kelt: Budapest, 2026. 05. 31.

Babai András

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Kelt: Budapest, 2026. 05. 31.

Török Regina

Abstract

Magyar

A villamosenergia-termelés jelentős változásokon ment keresztül az elmúlt évtizedekben. A hagyományos, szinkrongépes alapú termelést egyre inkább felváltják a megújuló energiaforrások (RES), mint például a naperőművek (PV) és a szélturbinás generátorok (WTG). Bár ezek a technológiák egyértelmű környezetvédelmi előnyöket nyújtanak, integrációjuk komoly kihívások elé állítja az átviteli hálózati rendszerirányítást. Bennük rejlő kiszámíthatatlanságuk és az inverteralapú energiaátalakítás a rendszer stabilitásának és megbízhatóságának csökkenéséhez vezet. Ezek a problémák még kritikusabbá válnak a nagy üzemzavarok vagy rendszerbomlások során, amelyek végső soron kiterjedt áramszüneteket, blackoutot is eredményezhetnek. Ezen kihívások kezeléséhez elengedhetetlen a nagy megújuló-résarányú villamosenergia-rendszerek viselkedésének megértése a különböző üzemállapotok mellett. Egy átfogóbb kutatási projekt részeként a hosszú távú cél egy olyan modell kifejlesztése, amely képes szimulálni a közép-európai szinkronrendszer viselkedését a rendszerbomlási események során.

Kutatásunk során az európai szinkronrendszerben fellépő rendszerbomlások szimulálására létrehozott modell fejlesztési folyamatán dolgoztunk. Ez a magába foglalja a teljes folyamatot az adatgyűjtéstől kezdve a PowerFactory modell felépítésén, valamint a PowerFactory API és az ENTSO-E Transparency Platform API használatán keresztül egészen a modellel végzett első szimulációkig.

English

Electric power generation has undergone significant changes in recent decades. Traditional synchronous generator-based production is increasingly being replaced by Renewable Energy Source (RES) such as photovoltaic (PV) power plants and wind turbine generators (WTGs). Although these technologies offer clear environmental benefits, their integration poses considerable challenges for transmission system operation. Their inherent unpredictability and inverter-based energy conversion lead to reduced system stability and reliability. These issues become even more critical during large disturbances or system separations, which may ultimately result in widespread blackouts. To address these challenges, it is essential to understand the behavior of power systems with high RES penetration under various operating scenarios. As part of a broader research project, the long-term goal is to develop a model capable of simulating the behavior of the Central European Synchronous Area during system splits.

This document presents the development process of the model built to simulate system separations in the European Synchronous Area. It covers the process from data collection through the construction of the PowerFactory model and the use of the PowerFactory API and the ENTSO-E Transparency Platform API, up to the first simulations performed with the model.

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Acronyms

AC alternating current. 4, 8

API Application Programming Interface. 14–17

AVR Automatic Voltage Regulator. 6

ENTSO-E European Network of Transmission System Operators for Electricity. 5, 6, 8, 13, 14

FFR Fast Frequency Response. 3, 7

GOV Turbine Governor. 6

PSS Power System Stabiliser. 6

PV photovoltaic. 3, 5

REEC Renewable Energy Electrical Control. 6

REGC Renewable Energy Generator/Converter. 6

REPC Renewable Energy Plant Control. 6

RMS Root Mean Square. 7, 17

RoCoF Rate of Change of Frequency. 2

TSO Transmission System Operator. 15

UCTE Union for the Co-ordination of Transmission of Electricity. 10

UFLS Underfrequency Load Shedding. 8

WECC Western Electricity Coordinating Council. 3, 6, 7

1 Introduction

To maintain operational security and continuity of supply in the European power system, an interconnected synchronous area is operated. The fundamental goal of this system is to enable cross-border electricity flows, ensuring mutual support among national grids during contingencies or power shortages. A key condition for the stable operation of the synchronous system is to continuously maintain the balance between electricity generation and consumption. Any imbalance results in an increase or decrease in system frequency.

This issue becomes particularly significant during system separation. Consider an operating state where a specific country imports electricity from an exporting region. If a system separation occurs at this moment, the previously balanced power flow is interrupted, creating a major power imbalance. Consequently, the system frequency can change drastically.

The significance of this problem is further increased by the declining share of synchronous generators in the power system, which reduces physical system inertia. The growing number of solar and wind power (1) plants connect to the grid predominantly through inverter-based generation units. The operational characteristics of inverter-based technology differ significantly from traditional synchronous systems. Therefore, analyzing the dynamic behavior of these units during system separation events is highly important.

Solar and wind power generation, 2005 to 2025

Electricity generation from solar and wind, measured in terawatt-hours¹.

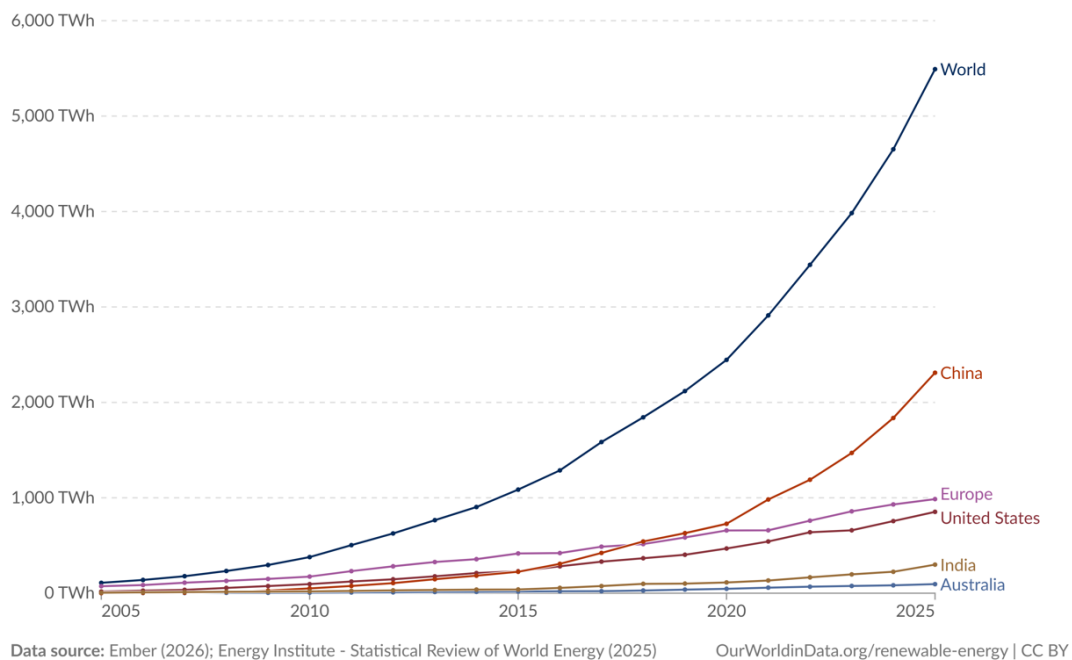


Figure 1 | Solar and wind power generation [1]

The main goal of our research is to investigate and evaluate potential European system separations occurring at national borders. Our study focuses on frequency, inertia, Rate of Change of Frequency (RoCoF), and inertia distribution.

To achieve this, it is necessary to create a grid model that represents the European synchronous system in a simplified form but with sufficient accuracy, while taking inverter-based generation units into account. An additional requirement for the model is to be suitable for fast calculations and dynamic studies.

1.1 Previous works

In our previous work, we investigated the European system split of January 8, 2021, where the continental power system separated into two independent regions [2]. To replicate this event, a simplified single-machine single-busbar network configuration was adopted based on a well-established model found in the literature [3].

Furthermore, significant focus was placed on the modeling and control of inverter-based resources. Initially, photovoltaic (PV) generation was integrated using the standard WECC model to understand its underlying control loops. The primary objective was to extend the baseline synchronous machine model of the 2021 system split by incorporating PV generation, thereby sharing the total generation between an aggregated synchronous machine and an aggregated PV system. This approach is highly relevant due to the rapid proliferation of solar energy; it allows us to analyze how the system would behave during a potential system split under high-penetration scenarios, such as a sunny noon in the summer of 2026. To enable active frequency support from the inverters, a synthetic inertia block was implemented alongside the built-in droop control. This modification successfully improved the frequency response following the system split, as power electronics can provide faster regulation than conventional synchronous generators. [2]

Following the PV studies, our research expanded into wind power modeling, specifically focusing on Type 3 doubly-fed induction generator wind turbines. This was also based on a Western Electricity Coordinating Council (WECC) model, which featured built-in Fast Frequency Response (FFR) controls.

Building upon the experience gained from modeling these individual dynamic components, the next step is to construct detailed, country-level aggregated models. By accurately representing the dynamic behavior of each country and their respective interconnections, the ultimate goal of this research is to develop a comprehensive, multi-area European power system model.

2 Model building in PowerFactory

The main goal of this research is to develop a European model that can accurately simulate system splits while remaining simple and computationally efficient. The model consists of two main components:

- **Single-node country models:** designed to represent the production and consumption characteristics of each individual country
- **Single-line cross-border connection models:** represent the physical parameters of the transmission lines connecting the countries

Single-node models play an important role in power grid simulations, especially in frequency stability studies. Their main advantage is that they can represent even very complex power systems using only a single node. This significantly reduces the number of variables and parameters that need to be considered, making the analysis simpler and more manageable.

The structure of the developed European model is shown in Figure 2.

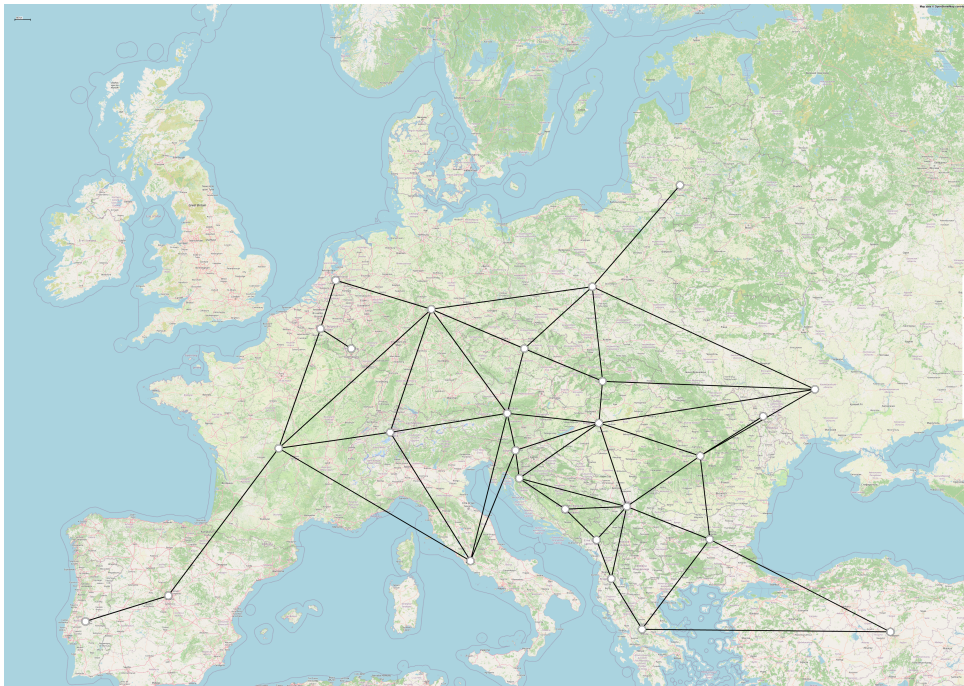


Figure 2 | European model

The model was developed using the DlgSILENT PowerFactory software.

2.1 Country models

Because this study focuses on frequency variations, only alternating current (AC) connections were included in the model, resulting in a system consisting of 27 countries. The countries considered in the analysis are shown in Figure 3.

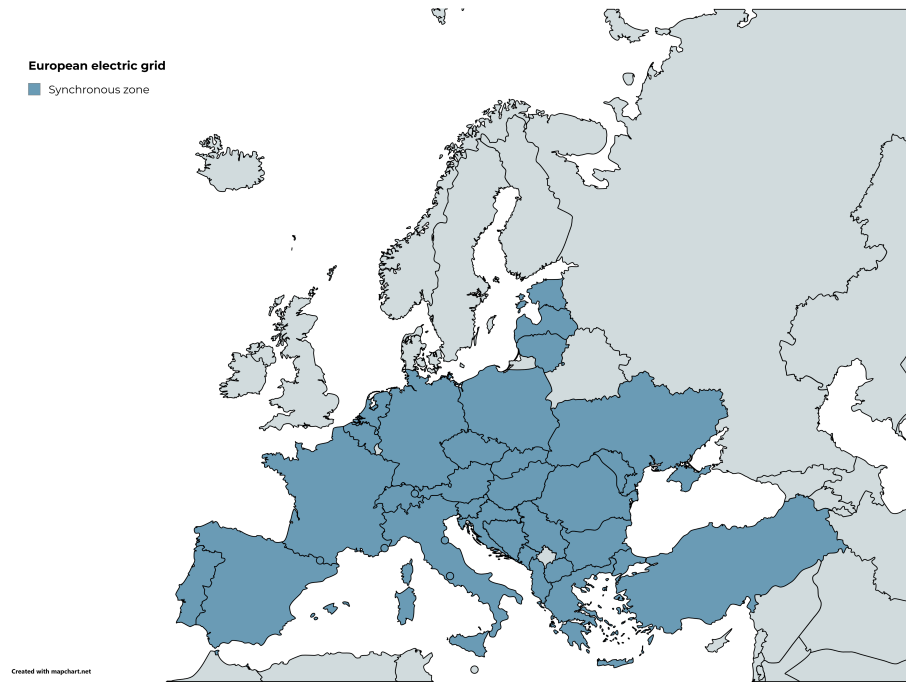


Figure 3 | Modeled countries

The country models consist of equivalent generators and equivalent loads. A distinction was made between synchronous generators, photovoltaic (PV) systems, and wind turbines due to their different technologies, in order to represent them more accurately. The equivalent representation means that each generation type aggregates the total installed capacity of the entire country into a single unit, while the load represents the country's total electricity consumption as a single equivalent load.

Thus, a single country model consists of a terminal that includes a synchronous generator, a PV system, a wind turbine, and a load. An example of the Hungarian model representation is shown in Figure 4.

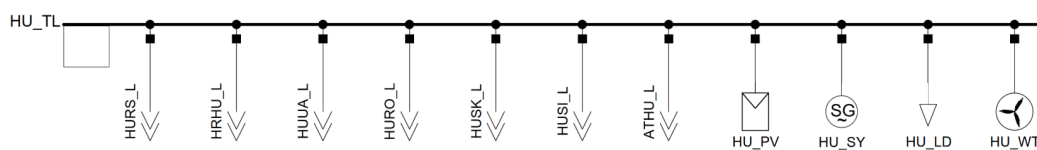


Figure 4 | Hungarian country model

We used the following naming convention shown in Table 1. The country codes refer to the two-letter country codes used by European Network of Transmission System Operators for Electricity (ENTSO-E), ensuring that all elements can be consistently identified.

Table 1 | Naming convention - Countries

Terminal	country_code_TL
Synchronous generator	country_code_SY
PV system	country_code_PV
Wind turbine	country_code_WT
Load	country_code_LD

2.1.1 Synchronous generator

The built-in synchronous machine model of the DigSILENT PowerFactory software was used in the investigations. The main parameters of a synchronous generator are the rated apparent power ($S_{synchronous}$), the active power output ($P_{synchronous}$), and the inertia constant (h). These parameters were set according to real-time measurement data, as detailed in Chapter 4.1.

The operation of the synchronous machine is influenced by several controllers, of which three were implemented in our model in accordance with the ENTSO-E modeling practice:

- Automatic Voltage Regulator (AVR) excitation controller: SEXS
- Power System Stabiliser (PSS) power system stabilizer: PSS2A
- Turbine Governor (GOV) turbine governor: TGOV1

Each synchronous generator uses a composite model for its controllers. This allows you to configure settings for each machine individually. All composite models are based on a common general type. Modifying this general type changes the control loops and other global settings for all generators at once.

Table 2 | Naming convention - Composite model 1

Synchronous generator	country_code_C_SY
-----------------------	-------------------

The German generator was selected as the slack generator for the European model due to its central location and strong interconnections with neighboring countries. Using a central generator as the reference ensures that network losses are distributed realistically across the grid, avoiding convergence issues. Additionally, since Germany has the largest power system capacity in Continental Europe, it serves as the most logical reference point for system stability and voltage angles.

2.1.2 PV system

To model the solar power plants, we used the *WECC Generic Model for Large-scale solar PV Plants* developed by Western Electricity Coordinating Council (WECC) [4]. The control system consists of three main parts:

- Renewable Energy Generator/Converter (REGC): provide current signals for network calculations
- Renewable Energy Electrical Control (REEC): for local active and reactive power control
- Renewable Energy Plant Control (REPC): for plant-level active and reactive power control

Our research focused specifically on the Renewable Energy Plant Control (REPC) controller because it manages the PV active power-frequency response, which is critical for inertia studies. We extended the model's frequency control by adding a synthetic inertia control block alongside the built-in droop control [2]. Notably, most actual PV units do not have droop control implemented.

In DigSILENT PowerFactory, we used the built-in static generator and configured it to be photovoltaic-based. Then, we used the model developed by the University of Kassel [5].

Similar to the synchronous generators, each PV system used a composite model containing the controllers, with all composite models linked to a single general type. A critical component of this

control structure is the added Synthetic Inertia loop, which allows the configuration of the inertia values and related parameters. The naming conventions for both the composite models and the inertia loops are summarized in Table 3.

Table 3 | Naming convention - Composite model 2

PV system	country_code_C_PV
Synthetic inertia loop	country_code_REPC_6_Synthetic Inertia

2.1.3 Wind turbine

In our model, the wind power plants were considered as Type 3 wind turbines. For this, we used the *Type 3 Wind Turbine Generator Model* developed by WECC [6]. The model also provides the option to simulate Fast Frequency Response (FFR).

Every wind turbine also has a composite model, and all of them are based on a single general type. The naming of these composite models is shown in Table 4.

Table 4 | Naming convention - Composite model 3

Wind turbine	country_code_C_WT
--------------	-------------------

2.1.4 Load

During the model development process, the built-in general load type was used to represent the total consumption of a country. In DIgSILENT PowerFactory Root Mean Square (RMS) simulations, three-phase loads can be represented as purely static, purely dynamic, or as a combination of both static and dynamic components [7]:

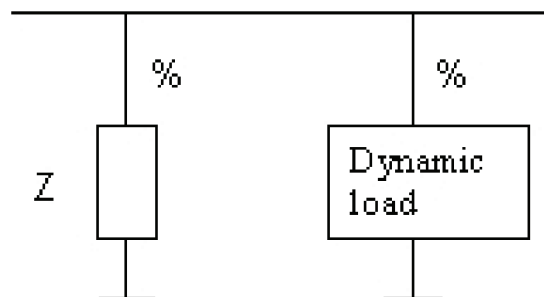


Figure 5 | Mixed Static-Dynamic Load Model [7]

The dynamic part of the loads can be represented as one of the following [7]:

- Linear
- Nonlinear voltage, linear frequency
- Nonlinear
- Nonlinear, regulated

We chose the *nonlinear voltage, linear frequency* setting. In this case, the voltage and frequency dependencies of the load can be defined using the following equations [7]:

$$P_d = P_0 \cdot \left[aP \cdot \left(\frac{|u|}{|u_{ini}|} \right)^2 + bP \cdot \frac{|u|}{|u_{ini}|} + cP \right] \cdot (1 + kpf \cdot \Delta f) \quad (2.1)$$

$$Q_d = Q_0 \cdot \left[aQ \cdot \left(\frac{|u|}{|u_{ini}|} \right)^2 + bQ \cdot \frac{|u|}{|u_{ini}|} + cQ \right] \cdot (1 + kqf \cdot \Delta f) \quad (2.2)$$

It is important to note that the settings are 0% static and 100% dynamic, since in this case, the load behaves 100% according to the defined dynamic equations.

For average load areas, $k_{pu} = 1$ and $k_{pf} = 1$, meaning a 1% change in voltage or frequency results in a 1% change in ΔP . For reactive power (Q), the combined effect of saturable iron-core inductors and capacitor compensation typically yields $k_{qu} = 3-8$ and $k_{qf} < 0$.

2.2 Power line models

The power lines represent the high-voltage AC interconnections between countries. We reduced the connections between any two countries to a single power line. The corresponding calculations can be found in Section 3.1.

In DlgSILENT PowerFactory, we used the following type for the power lines:

TYPE = N2XS(FL)2Y 1x100ORM/50 64/110kV

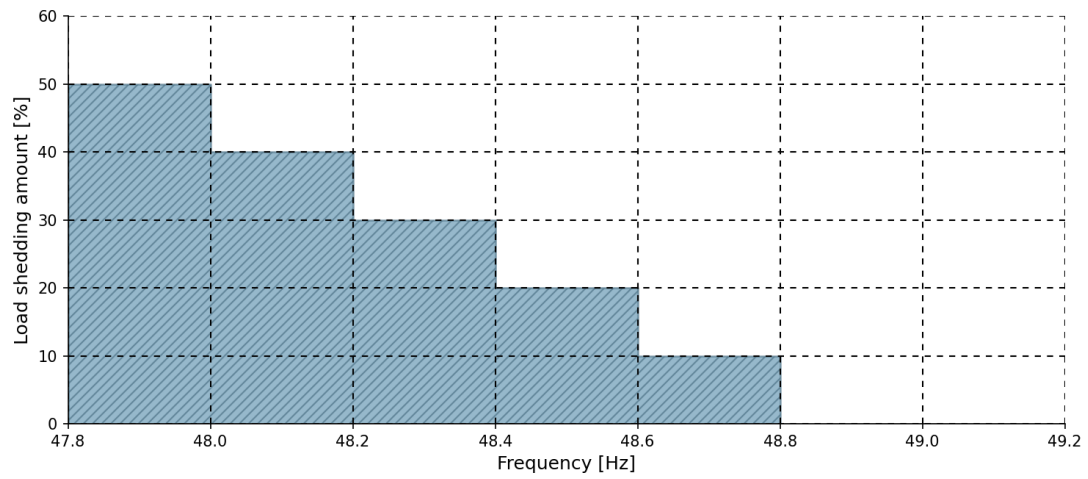
It was crucial that the power line models had many parameters that could be configured within their type. Therefore, for every power line, we defined a specific type called:

Table 5 | Naming convention - Powerlines

Powerline	country_code1country_code2
Powerline type	country_code1country_code2_T

2.3 Underfrequency load shedding

The model built in PowerFactory also contained underfrequency load shedding. This regulation logic was implemented with a relay at the connection point of every load. The relay logic followed a traditional six step ENTSO-E Underfrequency Load Shedding (UFLS) scheme [8], Graph 1. It is important to mention that this scheme is a general recommendation from ENTSO-E in practice, each country has its own UFLS scheme and regulation. In the future, for a more precise model, it would be advisable to set up each logic based on the current country UFLS scheme.



Graph 1 | Six step UFLS scheme according to ENTSO-E

3 Data collection

In order to parameterize the elements of the PowerFactory model, including transmission lines, generation units, and load units, the required data had to be collected from various sources. This section presents the data collection process and the methodology used to obtain and prepare the necessary model parameters.

3.1 Power line parameters

The parameters of the power lines were obtained from a non public ENTSO-E database. In the following section the processing of this database is presented in details.

3.1.1 The raw database

Original *.uct* file in Union for the Co-ordination of Transmission of Electricity (UCTE) format [9] The database contains data from different elements in the European power grid. The file structure is well defined, it contains the following information types (Table 6).

Table 6 | Network Data Categories [9]

Code	Description
C	Comments
N	Nodes
L	Lines
T	2 Windings Transformers
R	2 Windings Transformers Regulation
TT	2 Windings Transformers Special Description (optional)
E	Exchange Powers (optional)

In the center of our focus were the power lines, so the only data-block used from the database was the *L* section. In this block, each power line's parameters were collected in separate rows, where each character had its own meaning, defined in the UCTE format (Table 7).

Table 7 | Line Data Format Description [9]

Columns	Type	Length	Contents
1–8	character	8	Node 1 (code)
9	character	1	
10–17	character	8	Node 2 (code)
18	character	1	
19	character	1	Order code (1, 2, 3 ... 9, A, B, C ... Z)
20	character	1	
21	integer	1	Status (0, 1, 2 or 7, 8, 9)*
22	character	1	
23–28	real	6	Resistance R (Ω)
29	character	1	
30–35	real	6	Reactance X (Ω)
36	character	1	
37–44	real	8	Susceptance B (μS)
45	character	1	
46–51	integer	6	Current limit I (A)
52	character	1	
53–64	character	12	Element name (optional)
65	character	1	

The status flag has the following meanings:

- 0: real element in operation
- 8: real element out of operation
- 1: equivalent element in operation
- 9: equivalent element out of operation
- 2: busbar coupler in operation (definition: $R=0$, $X=0$, $B=0$)
- 7: busbar coupler out of operation (definition: $R=0$, $X=0$, $B=0$)

An example row from the UCT file:

```
MGYOR 21 XWI_GY21 0 4.955 25.629 165.2500 600.0
```

3.1.2 Required parameters

In order to configure the power line model mentioned in Section 2.2; the following parameters were needed (Table 8, 9).

Table 8 | Transmission line parameters

Parameter	Unit
Length	km
Max permissible current	kA

Table 9 | Transmission line type parameters

Parameter	Unit
Rated voltage	kV
Rated current	kA
AC-resistance (1,2 sequence)	Ω/km
Reactance (1,2 sequence)	Ω/km
Capacitance (1,2 sequence)	$\mu\text{F}/\text{km}$
Conductor material	—

3.1.3 Data processing

The processing of the data from the UCT database consisted of the following steps.

Reading the UCT file into a *pandas DataFrame*

Further data processing was carried out using the *pandas* [10] Python library, therefore the data was read into a *pandas DataFrame*. This creates an easy-to-handle and well-visualizable data structure.

Filtering the DataFrame

The *pandas DataFrame* contained all power lines from the European power grid; however, in this case, only the border-crossing power lines were required. Filtering these power lines was relatively straightforward, since the *Node 2* value of these lines was always a fictitious border node whose name started with the character *X*. Therefore, from the original *pandas DataFrame*, only the power lines for which the name of *Node 2* started with *X* were retained.

Merging the border-crossing lines

Since *Node 2* of the border-crossing lines was a fictitious node, every border-crossing power line was split into two different lines connected through one fictitious node (Table 10). Therefore, the lines sharing the same fictitious node had to be merged. During the merging process, the sum of parameters (R , X , B) was calculated, while in the case of the current (I), the smaller value was taken into account. As a result, a new merged power line was created (Table 11).

Table 10 | Split power line - example

Node 1	Node 2	STATUS	R [Ω]	X_L [Ω]	B [μS]	I_MAX [A]
MGYOR 21	XWI_GY21	O	4.955	25.629	165.2500	600.0
OWIEN 21	XWI_GY21	O	4.992	25.665	170.7449	600.0

Table 11 | Merged power line - example

Node 1	Node 2	Fictitious node	R [Ω]	X_L [Ω]	B [μS]	I_MAX [A]
MGYOR 21	OWIEN 21	XWI_GY21	9.947	51.294	83.976	600.0

Calculating the parallel resultant values

There are several cases in which a border contains multiple power lines, sometimes operating at different voltage levels (Table 12). In order to properly model each border connection using only a single equivalent power line, the resultant parameters of these power lines must be calculated.

Table 12 | Power lines on one border - example

Border	Node 1	Node 2	R [Ω]	X _L [Ω]	B [μ S]	I _{MAX}	U[kV]
ATHU	MGYOR 21	ONEUSI21	6.625	34.255	55.909	600.0	220
ATHU	MGYOR 21	OWIEN 21	9.947	51.294	83.976	600.0	220
ATHU	MGYOR 15	OZURND11	1.489	20.795	64.109	1999.0	380
ATHU	MSZHO 11	OZURND11	3.634	40.309	121.598	1999.0	380

In the first step, all values were converted to the nominal voltage level ($U = 380$ kV) based on Equations (3.1), (3.3), (3.2), and (3.4) (Table 13). The capacitive reactance was calculated from the capacitive susceptance. Furthermore, in this step, the capacitive reactance was converted into capacitance, since this parameter was required for the parametrization of the PowerFactory model.

$$R_{\text{common}} = R \left(\frac{380}{U} \right)^2 \quad (3.1)$$

$$X_{L,\text{common}} = X_L \left(\frac{380}{U} \right)^2 \quad (3.2)$$

$$X_{C,\text{common}} = \frac{1}{B} \left(\frac{380}{U} \right)^2 \quad (3.3)$$

$$I_{\text{common}} = I \left(\frac{U}{380} \right) \quad (3.4)$$

Table 13 | Power line parameters on the nominal voltage – example

Border	Node 1	Node 2	R _{com} [Ω]	X _{L,com} [Ω]	X _{C,com} [M Ω]	I _{com} [A]	C _{com} [μ F]
ATHU	MGYOR 21	ONEUSI21	19.765	102.199	0.053	347.368	0.060
ATHU	MGYOR 21	OWIEN 21	29.677	153.034	0.036	347.368	0.090
ATHU	MGYOR 15	OZURND11	1.489	20.795	0.016	1999.000	0.204
ATHU	MSZHO 11	OZURND11	3.634	40.309	0.008	1999.000	0.387

In the second step the parallel resultant values were calculated according to rules of the parallel resultant value calculation (Table 14).

Table 14 | Equivalent border parameters - example

Border	R [Ω]	X _L [Ω]	C [μ F]	I [A]
ATHU	0.4849	5.6044	1.4807	4692.74

3.2 Generation and load data

3.2.1 ENTSO-E Transparency Platform and API

As part of its data publication system, ENTSO-E created the so-called *Transparency Platform*. Numerous historical and forecast datasets from the European electrical grid are published on this platform [11]. In this case, the only relevant data were the historical *generation* and *load* data for

each modelled country, which were usually published with a fifteen-minute resolution, except for (...), where only hourly resolution was available.

In order to efficiently retrieve the required data, ENTSO-E developed an API for the Transparency Platform [12]. Based on this API, a Python library called *entsoe-py* was developed [13], which was also used in the present research.

During the data retrieval process, the generation types shown in Table 15 were requested for each country [14].

Table 15 | ENTSO-E production type codes

Code	Production Type
B01	Biomass
B02	Fossil Brown coal/Lignite
B03	Fossil Coal-derived gas
B04	Fossil Gas
B05	Fossil Hard coal
B06	Fossil Oil
B07	Fossil Oil shale
B08	Fossil Peat
B09	Geothermal
B10	Hydro Pumped Storage
B11	Hydro Run-of-river and poundage
B12	Hydro Water Reservoir
B13	Marine
B14	Nuclear
B15	Other renewable
B16	Solar
B17	Waste
B18	Wind Offshore
B19	Wind Onshore
B20	Other
B25	Energy storage

3.2.2 Data processing

Once the data was fetched, these production types were merged into three main production types, wind (*B18, B19*), solar (*B16, B25*) and synchronous generation (*the rest*). These three generation types were previously built into the PowerFactory model.

From the load side in the Transparency Platform there was only one value, so there was no further processing needed.

Since the model only contains the countries showed in Fig. 3; there was a slight inequality between the generation and the load. In order to equalize it and to start our model from an equalized initial condition, the load value of each country was modified (increased or decreased) based on their initial load contribution (Eq. (3.5)).

$$P'_{load,i} = P_{load,i} - \left(\sum P_{load,i} - \sum P_{gen,i} \right) \frac{P_{load,i}}{\sum P_{load,i}} \quad (3.5)$$

3.2.3 Handling missing data

About the generation and load values of some countries, the Transparency Platform wasn't able to provide any data (*Turkey*), or it just partially had any data (*Albania, Ukraine*). In order to overcome this issue the following steps were implemented.

Turkey

From the Turkish electric grid there is no data available on the Transparency Platform, neither generation nor load data. In order to address this issue, the required data was obtained from the *EPIAS Transparency Platform* [15]. Until the time of the creation of this document, the missing data was obtained manually, however the EPIAS also offers an API solution, which could be implemented into the current project in order to facilitate and fasten the data fetching process.

Ukraine

From the Ukrainian electric grid there is no data available on the Transparency Platform, neither generation nor load data. During our study cases, until now we only examined network disturbances which happened before the synchronisation of the Ukrainian and Moldavian with the Continental European grid in 16 of March 2022 [16]. Thus there were no data required from the Ukrainian power system. In order to perform simulations after that date with real grid parameters, an additional data source should be researched and implemented into the model.

Albania

From the data from the Albanian electric grid the only missing value are the generation values. After a thorough research, the only platform found was the official website of the Albanian Transmission System Operator (TSO) [17]. Since there was no API developed for this platform, the only way to access this data is to manually search for the required time period on the website of the Albanian TSO.

3.3 Extreme value finder

During the data fetching process, generation and load values can be retrieved for a selected time interval (e.g., 2026.01.01–2026.01.05). However, only a single instantaneous operating point is required for the parametrization of the model. To select one representative timestamp from the available dataset, an extreme value finder function was implemented. This function identifies extreme operating conditions within the specified time interval by evaluating solar and/or wind and/or synchronous generation values, supporting both maximum and minimum searches. Thus, distinct extreme cases can be examined.

4 PowerFactory API

Once all the necessary data was obtained, the next step is the parametrization and the interaction with the PowerFactory model. With such a large amount of data, the easiest way to do that is via the PowerFactory API. For this API, PowerFactory offers a closed source python library [18], which was used for the current project as well.

4.1 Parametrization of the PowerFactory model

The elements of the PowerFactory model, e.g., generators, loads, regulators, switchgear, and power lines, were parameterized using the PowerFactory API. This method provides a faster and more reliable parameter setup process compared to manual configuration in PowerFactory. The main model parameters are presented in Table 16.

Table 16 | PowerFactory model parameters

Code	Dimension	Value/calculation/name
V	kV	380
$P_{synchronous}$	MW	based on the ENTSO-E database
$S_{synchronous}$	MW	$1.2 \cdot P_{synchronous}$
P_{PV}	MW	based on the ENTSO-E database
S_{PV}	MW	P_{PV} or in case of de loaded operation [2]: $P_{PV} \cdot PV_{DL}$
P_{WT}	MW	based on the ENTSO-E database
h	s	inertia constant
λ	$\frac{MW}{Hz}$	power-frequency characteristic
σ	p.u.	aggregated droop

4.1.1 Inertia constant

The inertia constant should be calculated for each country. For this purpose, the formula presented in [19] was used (Equation (4.1)), where H is a fixed inertia constant with a value of 4s.

$$h_i = H \frac{P_{synchronous}}{P_{synchronous} \cdot P_{PV} \cdot P_{WT}} \quad (4.1)$$

4.1.2 Power-frequency characteristic

The calculation of the power-frequency characteristic was performed by the formula proposed in [3]. Based on that formula, the power-frequency characteristic of the Central European interconnected area can be calculated as presented in (Equation (4.2) - (4.5)). It is important to note that the effect of loads was neglected in the current calculation, since the model already contains sufficient loads to represent their impact.

$$\lambda_{min} = \frac{3000 \text{ MW}}{0.2 \text{ Hz}} \quad (4.2)$$

$$\lambda_{add} = 30\% \lambda_{min} \quad (4.3)$$

$$\lambda_{surplus} = \frac{50\%}{50 \text{ Hz}} P_{gen} \quad (4.4)$$

$$\lambda^* = \lambda_{min} + \lambda_{add} + \lambda_{surplus} \quad (4.5)$$

The power–frequency characteristic values for each country can be calculated from λ^* according to Equation (4.6), where c_i represents the contribution factor of the given country. These factors should be calculated separately for each scenario; however, for the system split of 8 January 2021, they were also presented in [3].

$$\lambda_i = \lambda^* \cdot c_i \quad (4.6)$$

4.1.3 Aggregated droop

The value of aggregated droop is another important parameter of the synchronous machine model. The value of it was calculated for each country in the model, based on the formula presented in [3].

$$\sigma_i = \frac{1}{\lambda_i} \frac{P_n}{f_n} \quad (4.7)$$

Where P_n is the nominal power and f_n is the nominal frequency of the actual country.

4.2 Simulations in PowerFactory

The simulations in PowerFactory were also run with the PowerFactory API. Two types of simulations were performed, a general *load flow* simulation and time variant *RMS* simulations. The simulation events for the RMS simulation were also switchable through the API access. The results of the RMS simulations, for easier post processing, were saved as a *csv* file.

5 Further studies

5.1 Conclusion

The original goal of the research project was to build up a simple yet accurate dynamic model in order to examine the effect of system splits on the frequency stability of the Central European electric grid. This model was successfully built up and it is ready for running RMS simulations. The results of these simulations can later be used to assess the frequency response of the system under various split scenarios, identify critical operating conditions, evaluate the adequacy of frequency containment measures, and support the development of strategies aimed at improving the resilience and stability of the interconnected power system.

5.2 Future Plans

The next milestone of the current research project is to validate the developed model. To achieve this, a real-world system split event will be reproduced and simulated, and the model response will be compared with available measurements and historical data.

Following the validation process, a series of system separation scenarios will be investigated. These scenarios will consider different generation-load balances, generation ratios, and inertia distributions across the separated areas. The resulting simulations will enable a systematic assessment of their impact on frequency dynamics and overall system stability, providing valuable insights into the resilience of the Central European power system under split conditions.

6 Felhasznált irodalom

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